

**COURSE NAME & CODE:** ECA 0303 – Signals and systems

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## **Experiment 1.2**

### **Generation of Discrete Time Signals**

#### **Aim**

To generate and verify a discrete-time unit impulse signal using MATLAB. The Aim is to ensure that the signal generator can accurately produce an impulse signal at a specified time index, which is crucial for analyzing and understanding the behavior of discrete-time systems.

#### **Program:**

```
% Generation of Discrete Time Signal
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Discrete-time index
```

```
n = -10:10;
```

```
% Discrete-time sinusoidal signal
```

```
x = sin(0.2*pi*n);
```

```
% Plotting
```

```
figure;
```

```
stem(n, x, 'filled');
```

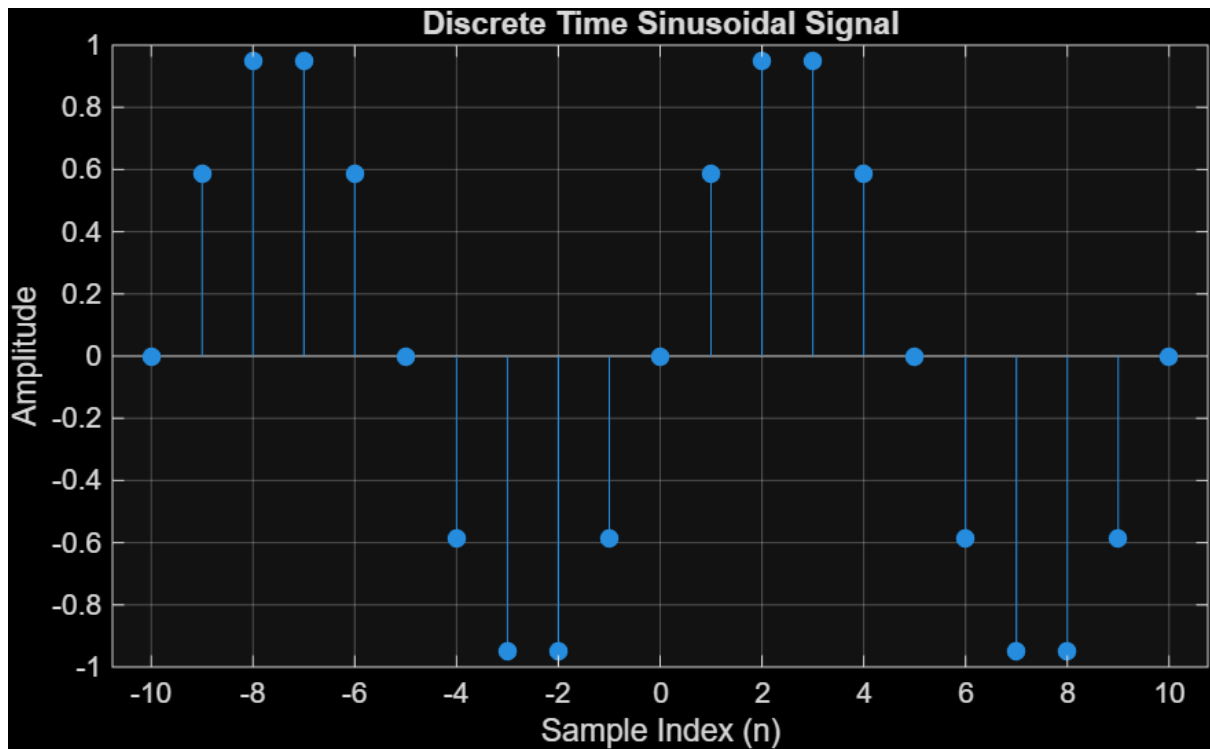
```
grid on;
```

```
title('Discrete Time Sinusoidal Signal');
```

```
xlabel('Sample Index (n)');
```

```
ylabel('Amplitude');
```

### Output Waveform:



### Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete time systems.